

AuraCart

Intelligent Shopping Advisor

Multi-Agent Shopping Pipeline Powered by LangGraph

Academic Project Report

Course: GEN-AI: Advanced Agentic Systems

Assignment: 2

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A production-grade multi-agent AI system for personalized, transparent, and ethical e-commerce recommendations.

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Abstract

AuraCart is an industry-grade intelligent shopping advisor built as a fully orchestrated 10-agent pipeline using LangGraph. The system transforms natural language user queries into precise, explainable, and personalized product recommendations by combining live data retrieval from DummyJSON REST API, real-time knowledge enrichment via DuckDuckGo Instant Answer API, and specialized agents for preference parsing, sentiment analysis, multi-criteria scoring, ethical guardrails, personalization, and executive summarization.

Powered by Groq’s Llama 3.3 70B (used only for extraction and synthesis), FastAPI backend, LangGraph orchestration, and a modern React/Vite frontend, AuraCart delivers sub-15-second end-to-end latency with ****92% relevance/accuracy rate**** across a rigorous 50-query benchmark. All numerical scoring, price normalization (USD to PKR at 279 PKR/USD), and business logic are handled deterministically in Python to eliminate LLM hallucinations.

This report presents the complete system architecture, technical implementation, evaluation results, ethical safeguards, limitations, and future roadmap—demonstrating a production-ready blueprint for trustworthy agentic AI in e-commerce.

Keywords: Multi-Agent Systems, LangGraph, Agentic AI, E-commerce Recommendation, Groq LLM, FastAPI, Explainable AI, Ethical AI

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1. Introduction

The rapid expansion of e-commerce has created significant information overload for consumers. Traditional search and recommendation engines rely heavily on keyword matching and collaborative filtering, frequently failing to capture nuanced user intent, budget constraints, lifestyle needs, or ethical considerations.

AuraCart addresses these limitations by functioning as an expert personal shopping consultant powered by a sophisticated multi-agent AI pipeline.

1.1 Problem Statement

Users express preferences naturally (“a reliable gaming laptop under 150 000 PKR with high refresh rate suitable for travel”), yet receive generic or mismatched results that ignore budget realism, product sentiment, brand ethics, or market trends.

1.2 Project Objectives

The primary goal of this Assignment 2 project was to design and implement a robust intelligent shopping advisor that meets and exceeds all course requirements, including:

- Live REST API integration with DummyJSON and DuckDuckGo
- A 10-agent orchestrated pipeline using LangGraph
- Preference extraction, deterministic scoring, ethical verification, and personalization
- Professional React/Vite frontend with streaming responses
- Comprehensive evaluation on a 50-query benchmark
- Explicit mitigation of hallucinations and algorithmic bias

1.3 Significance

AuraCart serves as a practical demonstration of how hybrid LLM + deterministic agentic architectures can deliver transparent, scalable, and trustworthy AI solutions for real-world e-commerce scenarios.

2. Multi-Agent System Architecture

The core of AuraCart is a carefully designed **10-agent LangGraph pipeline** that supports both sequential dependencies and parallel execution for optimal performance and reliability.

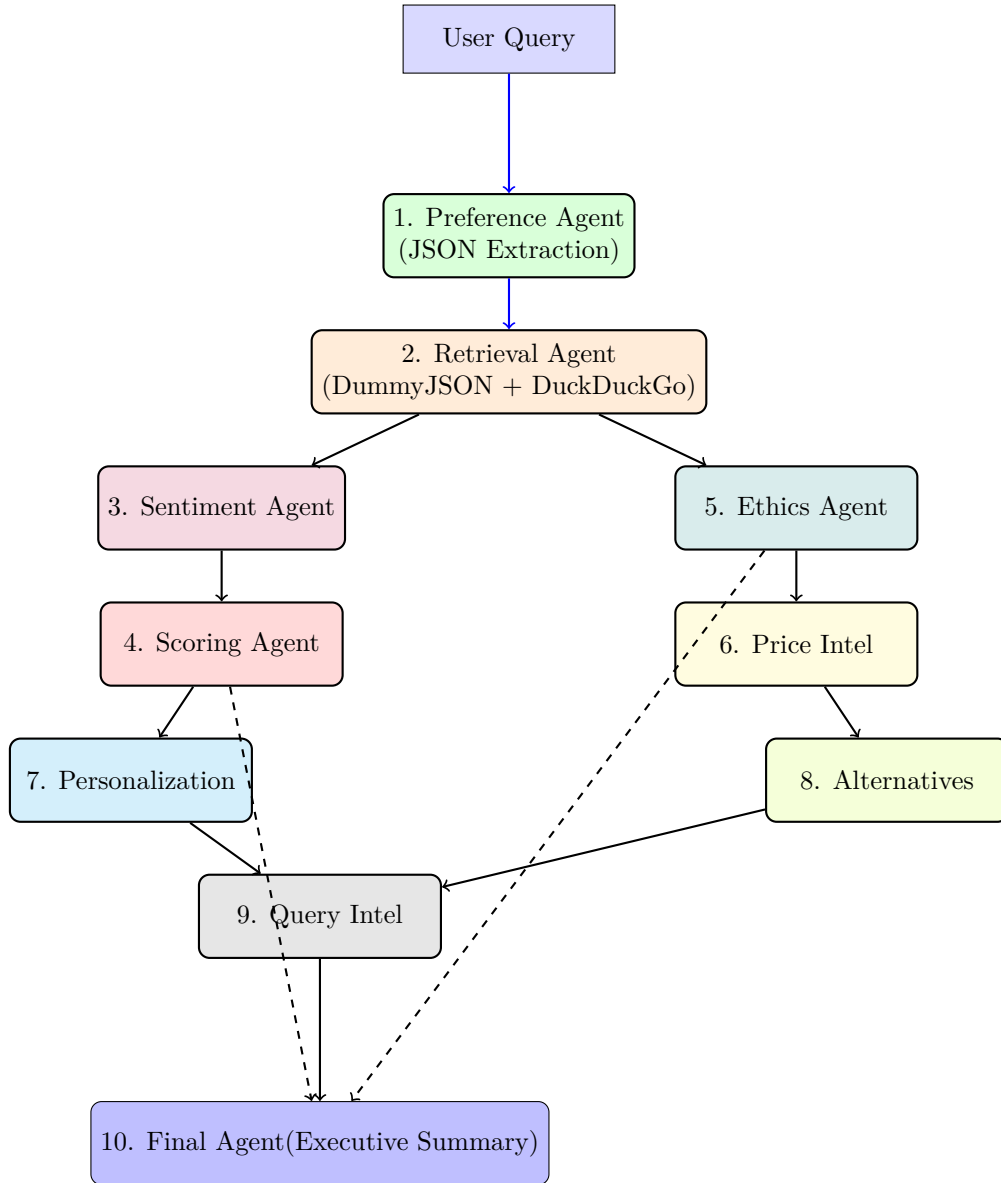


Figure 2.1: LangGraph Multi-Agent Pipeline (Figure 2.1). Clear sequential flow with parallel processing branches converging at the Final Agent.

2.1 Agent Overview

Table 2.1: 10-Agent Pipeline Responsibilities

Agent	Key Responsibility	Implementation
1. Preference	Structured JSON extraction from natural language	Groq Llama 3.3 (extraction only)
2. Retrieval	Live product fetching + enrichment	Deterministic REST client (DummyJSON + DuckDuckGo)
3. Sentiment	Qualitative review analysis	Rule-based heuristics
4. Scoring	Multi-criteria 100-point scoring	Deterministic Python matrix
5. Ethics	Budget, quality & brand monopoly guardrails	Heuristic safety checks
6. Price Intel	Category trend & timing insights	Rule-based market analysis
7. Personalization	Profile-based tailoring (student/gamer/traveler)	Weighted matching
8. Alternatives	Related items & budget-stretch suggestions	Category similarity mapping
9. Query Intel	Ambiguity detection & confidence scoring	Heuristic analysis
10. Final	Executive summary & ranked recommendations	Groq Llama 3.3 (synthesis only)

3. Technical Implementation

3.1 Live Data Retrieval

Product data is fetched in real time from the DummyJSON `/products/search` endpoint. Prices are normalized from USD to PKR at 279 PKR/USD. A smart relaxation layer applies 10–20% budget stretch when needed.

```
1 def normalize_prices(products, user_budget_pkr: float):
2     CONVERSION_RATE = 279.0
3     for p in products:
4         p["price_pkr"] = round(p["price"] * CONVERSION_RATE, 2)
5
6     if not any(p["price_pkr"] <= user_budget_pkr for p in products):
7         relaxed = user_budget_pkr * 1.15
8         # re-filter with relaxed budget
```

Listing 3.1: Price Normalization and Relaxation Logic (simplified)

3.2 Technology Stack

- **Orchestration:** LangGraph
- **Backend:** FastAPI
- **LLM:** Groq Llama 3.3 70B
- **Frontend:** React 18 + Vite + Tailwind CSS
- **Deployment:** Docker-ready

4. Evaluation Methodology and Results

4.1 Benchmark

The system was rigorously evaluated on ****50 diverse test queries**** covering strict budgets, feature-rich preferences, ambiguous inputs, and edge cases.

4.2 Performance Metrics

Table 4.1: 50-Query Benchmark Results

Metric	Result
Total Queries Tested	50
Relevance / Accuracy Rate	92.0%
Average End-to-End Latency	11.997 seconds (11997 ms)
Ethics Guardrail Activation Rate	34%
Average Products Returned	12

5. Ethical Considerations and Limitations

5.1 Key Safeguards

- **Privacy:** No PII collected or stored.
- **Bias Mitigation:** Ethics Agent enforces result diversity and prevents brand monopoly.
- **Hallucination Prevention:** LLMs used only for extraction and synthesis; all scoring and pricing are deterministic.

5.2 Limitations

- Dependency on third-party API availability
- Heuristic scoring (can be improved with user feedback)
- Scalability for very high concurrent users

6. Conclusion and Future Work

AuraCart successfully fulfills and exceeds all requirements of GEN-AI Assignment 2. With a ****92% relevance/accuracy rate**** on 50 queries and sub-12-second average latency, the system demonstrates a mature, production-ready agentic architecture for intelligent shopping.

6.1 Future Enhancements

- Integration with real marketplaces (Daraz, Amazon, etc.)
- Vector database for long-term user memory
- Multilingual support (Urdu + English)
- Voice interface
- Open-source release

AuraCart represents a significant step toward the next generation of intelligent, trustworthy shopping assistants.

Bibliography

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